

Gaoxiang Zhao

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Education

University of Pennsylvania

August 2025 – May 2027

MSE in Robotics

Philadelphia, USA

- Related Courses: Machine Learning (A), Learning in Robotics (A), Physical Intelligence (A)

Wuhan University

September 2021 – July 2025

BEng in Communication Engineering

Wuhan, China

Experience

Tencent

June 2026 – August 2026

Research Intern (Advisor: Kai Yan)

Shenzhen, China

- Worked on agentic 3D scene generation and 3D asset generation.

GRASP Lab

December 2025 – Present

Research Intern (Advisor: Kostas Daniilidis)

Philadelphia, USA

- Working on event-based vision and its applications in robotics.

Manycore Tech

August 2024 – November 2024

Research Intern

Hangzhou, China

- Developed Monte-Carlo physically-based rendering algorithms for a production-scale rendering pipeline supporting photorealistic synthetic data generation.

Selected Projects

Sim2Real Drone Racing with Deep RL (ESE 6510)

February 2026 – April 2026

- Implemented a simplified version of *Champion-level drone racing using deep reinforcement learning* (Kaufmann et al., Nature 2023) as a course project.
- Trained a drone (Crazyflie 2.1) racing policy with PPO in Isaac Lab (8,192 parallel environments) on a 3-lap, 7-gate indoor course, using a 36D body-frame observation (velocity, rotation matrix, gate corners) and a 10-term reward function.
- Transferred the policy to hardware without real-world fine-tuning using dynamics domain randomization (thrust-to-weight, drag, PID gains) and three iteration-conditioned curricula (approach distance, DR magnitude, speed reward weighting); the real drone completed 3 laps in 20s.

Physically-based Renderer

December 2023 – January 2025

- Developed a physically-based rendering engine in C++ with support for diverse geometric primitives, materials, textures, and light sources.
- Implemented path tracing, bidirectional path tracing, and photon mapping for light transport simulation in complex scenes.

Diffraction Simulation

June 2024 – August 2024

- Implemented the *Free-Space Diffraction BSDF* (Steinberg et al., 2024) in the PBRT-v3 renderer, extending Monte Carlo path tracing with edge-based Fraunhofer diffraction.
- Added a spectral rendering mode on top of the original monochromatic and RGB tricolor modes, and studied how spectral sampling affects convergence.

Technical Skills

- Programming: C++, Python, MATLAB
- Tools: $\LaTeX$, Linux, Git, Slurm, PyTorch, ROS, Unity, OpenGL, Isaac Lab, MuJoCo
- Language: English (Proficient), Mandarin (Native)